# RepSViT: An Efficient Vision Transformer Based on Spiking Neural Networks for Object Recognition in Satellite On-Orbit Remote Sensing Images

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***Abstract*— The role of on-orbit computing for satellites is transitioning from being a backup measure to becoming a primary key function. However, the limited computing resources available on satellites make it difficult to deploy advanced models with large parameters. Additionally, satellite on-orbit computing requires high speed and accuracy, posing significant challenges for developing suitable models. To overcome these challenges, we propose an efficient vision transformer, RepSViT, for satellite on-orbit computing. The RepSViT introduces spiking neural networks (SNNs) with high biological plausibility, event-driven property, and low power consumption into the field of remote sensing image processing and satellite on-orbit computing for the first time and incorporates structural reparameterization. Specifically, we design a dynamic dilated spiking convolution ($D^2SC$) based on SNNs to improve the feature extraction capability and efficiency of RepSViT. We also develop a spiking guided attention module (SGAM) to make RepSViT pay more attention to object-related features with lower computational costs. Furthermore, we design an efficient coupled fine-coarse-grained block (ECFC) to enhance the model's capability in extracting coarse and fine-grained features. To ensure effective feature extraction, inference speed, and reduced computational costs, we design a reparameterized feed-forward network (RepFFN). RepSViT achieves an inference latency of 8.33 ms and a recognition accuracy of 95% on an embedded GPU, utilizing 3.77 million parameters and consuming 0.6 GFLOPs computational costs.**

***Index Terms*— Fine-grained object recognition, remote sensing image (RSI), satellite-on-orbit computing, spiking neural networks (SNNs).**

## I. Introduction

AS THE satellite industry continues to integrate cutting-edge technologies [1], the role of on-orbit computing is transitioning from a backup measure to a primary critical function, with the potential to become a standard feature of advanced satellites in the future. However, there are currently numerous challenges associated with implementing on-orbit computing tasks for satellites [2].

First, satellite on-orbit computing tasks impose stringent requirements on the accuracy of computational models [3]. Transformers have achieved tremendous success not only in natural language processing tasks but also in computer vision tasks, such as object detection [4], [5], object recognition, semantic segmentation, and others [6]. This success has given rise to various variants of vision transformers (ViTs) [7]. Fu et al. [8] proposed progressive learning ViT for open set recognition of fine-grained objects in remote sensing images (RSIs). Liu et al. [9] proposed dual-branch transformers for RSI change captioning. Xu et al. [10] proposed an RSSFormer based on a transformer for remote sensing land-cover segmentation. Mo et al. [11] proposed swin-conv-dspp and global local transformer for RSI semantic segmentation. Xu et al. [12] used ViT as a teacher model to guide a small network for RSI scene classification. Xu et al. [13] proposed cross spatial–spectral dense transformer for hyperspectral image classification. Zhou et al. [14] proposed an arbitrarily oriented SAR ship detector with pyramid ViT. In the aforementioned studies, whether in the context of object detection [15], object recognition, semantic segmentation, or scene classification [16], enhanced ViT models have demonstrated competitive performance. However, these models require substantial computational resources to operate, rendering them impractical for deployment on resource-constrained satellites. One of the challenges addressed in this article is how to improve and apply the high-performance ViT to on-orbit object recognition tasks while preserving its exceptional performance.

Furthermore, the computational power available for on-orbit computing models is limited by satellites [17]. Some researchers have recognized the significant computational costs associated with ViT as a primary challenge in deploying it on edge devices. Apple [18] introduced structural reparameterization techniques into ViT models, proposing FastViT, which effectively improves inference speed while reducing computational costs. Huang et al. [19] introduced a globally efficient aggregation scheme into ViT, proposing lightViT, which achieves a better balance between accuracy

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and efficiency on pure transformer blocks without convolution. Pan et al. [20] developed EdgeViTs, a lightweight model based on optimal integration of self-attention and convolution, utilizing a cost-effective local–global–local information exchange bottleneck. These aforementioned studies have achieved outstanding performance on natural images with high color saturation and rich object information, but their performance degrades on RSIs where object features are more difficult to extract compared with natural images. Uneven resolution changes caused by tilt imaging and various interferences lead to distortion in RSIs. Additionally, the large-scale scenes and top-down perspective characteristics of RSIs make it even more challenging to extract object features compared with natural images. Therefore, the aforementioned methods may not perform optimally when applied to RSI object recognition. The investigation of lightweight ViT models for RSI object recognition is still an area of ongoing research. Designing lightweight ViT models suitable for on-orbit satellite computing is another challenge addressed in this article.

Finally, satellite on-orbit computing tasks place high demands on the inference speed of models [21]. Time-critical applications and services, such as firefighting and earthquake relief, require higher inference speeds for on-orbit computing models. SNNs provide a promising and energy-efficient solution for on-orbit computing models due to their high biological plausibility, event-driven property, and low power consumption [22]. One of the challenges addressed in this article is how to incorporate SNNs into on-orbit computing models and harness their energy-efficient advantages.

To address the above-mentioned challenges, we propose an efficient ViT based on SNNs using structural reparameterization for satellite on-orbit computing, RepSViT. Our main contributions are summarized as follows.

1) We propose an efficient ViT, RepSViT, for satellite on-orbit computing. The RepSViT introduces SNNs into the field of satellite on-orbit computing for the first time and incorporates structural reparameterization techniques.
2) We design a dynamic dilated spiking convolution ($D^2SC$) based on SNNs to enhance both the feature extraction capability and efficiency of RepSViT. Additionally, we develop an efficient coupled fine-coarse-grained (ECFC) block to strengthen the model's ability to extract features at both coarse and fine-grained levels.
3) We design a spiking guided attention module (SGAM) based on $D^2SC$ to make RepSViT pay more attention to object-related features with lower computational costs. Furthermore, we develop a reparameterized feed-forward network (RepFFN) to ensure the model's feature extraction capability and inference speed while reducing computational costs.

The rest of this article is organized as follows. Section I introduces the related work to the RepSViT. Section II details our proposed RepSViT. Section III demonstrates the experimental results on the ground server and the satellite board to evaluate the effectiveness of the proposed approach. Finally, Section IV draws conclusions.

## II. Related Work

To meet the requirements of "lightweight, fast, and accurate" on-orbit computing for satellites, we propose an efficient ViT, RepSViT, that leverages the strengths of both SNNs and deep neural networks (DNNs) through their integration. So in this section, we review the related work from three aspects: 1) lightweight artificial neural networks; 2) efficient spiking neural networks (SNNs); and 3) hybrid neural networks combining SNNs and DNNs.

### A. Lightweight Artificial Neural Networks

To enable advanced artificial neural networks to be deployed on edge devices, researchers have devised numerous lightweight artificial neural networks [23]. The design approaches for these lightweight networks can generally be categorized as follows: structural optimization design [24], weight quantization [25], knowledge distillation [26], network pruning [27], and structural reparameterization [28]. Li et al. [29] used random topology and random multiscale mapping to improve neural architecture search (NAS) and proposed an automated design of multiscale and lightweight neural network for remote-sensing image recognition; Mei et al. [30] proposed a new step activation quantization method for RSIs classification, which constrains the input of CNN's network layer so that the data can be represented by low-order integers; Yang et al. [31] proposed adaptive knowledge distillation for lightweight remote sensing object detectors optimizing; and Lu et al. [32] proposed an energy-based filter pruning framework to reduce the size of the original model for satellite on-orbit computing. Although lightweight artificial neural networks created using these approaches have shown promising performance, they ultimately still rely on computationally intensive floating-point multiplication operations due to the underlying complexity of artificial neural networks.

### B. Energy-Efficient Spiking Neural Networks

SNNs, as the third generation of artificial neural networks [33], are designed with significant inspiration from biological neural networks [34]. Brain-inspired SNNs offer notable benefits in terms of energy efficiency and rapid inference computations when deployed on neuromorphic hardware platforms like TrueNorth [35] and Loihi [36]. These advantages stem from the utilization of a biologically inspired binary spike transmission mechanism, which enables the networks to circumvent the need for multiplicative operations during inference [37]. With their high biological plausibility, event-driven property, and low power consumption, SNNs offer a promising energy-efficient alternative for edge devices [38]. However, the binary coding characteristic of SNNs makes the spike generation process nondifferentiable, resulting in the problem of low training accuracy [39]. Researchers have explored surrogate gradient methods to alleviate the training difficulties of SNNs [40]. To solve the problem of models stuck in local optima, Che et al. [41] introduced the differential approach to surrogate gradient search where the surrogate gradient function is efficiently optimized locally. However,

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the gradient descent with surrogate gradient does not match with the loss landscape in SNNs and is easy to get trapped in a local minimum with low generalizability. Although using suitable optimizers and weight decay helps ease this problem, in general, the accuracy of SNNs still falls behind that of DNNs.

### C. Hybrid Neural Networks Combining SNNs and DNNs

The high performance of DNNs comes at the cost of extensive floating-point multiplication operations, while SNNs are known for their energy efficiency and low power consumption but exhibit moderate performance [42]. Consequently, there has been significant interest among researchers in devising hybrid neural networks that combine the advantages of both pulse-based and artificial neural networks [43]. The design of such hybrid neural networks, which integrate SNNs and DNNs, has attracted substantial attention. These networks retain the high-performance capabilities of DNNs while harnessing the energy efficiency and computational efficacy of SNNs [44]. Therefore, hybrid neural networks serve as a bridge for implementing artificial intelligence on edge devices [45]. However, current research on hybrid neural networks primarily focuses on terrestrial applications, leaving a significant research gap concerning their application to satellite-based computational tasks. Therefore, it is imperative to address the research on hybrid neural networks tailored specifically for satellite-based computational tasks.

## III. Methodology

RepSViT is an object recognition model designed for on-orbit computation in satellites. This model introduces SNNs to the field of satellite on-orbit computation for the first time, along with the utilization of structural reparameterization techniques. SNN, known for its high biological plausibility, event-driven property, and low power consumption, offers a promising energy-efficient alternative for satellite on-orbit computation. Specifically, we have devised the $D^2SC$ based on SNN to enhance model efficiency. We have also developed the SGAM to expand the receptive field for feature extraction and reduce energy consumption. Moreover, the ECFC block has been designed to augment the model's ability to extract fine-grained and coarse-grained features. Additionally, the RepFFN has been devised to ensure the model's feature extraction capability, inference speed, and energy efficiency. RepSViT achieves a tradeoff between object recognition speed and accuracy under extremely limited computing resources in satellite environments.

In this section, we will provide a detailed explanation of the design principles of RepSViT, the formula derivation of related lightweight networks, and the processing workflow for satellite on-orbit object recognition based on RepSViT. We will primarily focus on four aspects: 1) RepSViT network architecture, which includes a complex training structure with strong feature extraction capabilities and a concise inference structure for fast and energy-efficient processing; 2) $D^2SC$, a novel efficient convolution employed in RepSViT; 3) ECFC feature extraction method; and 4) satellite on-orbit object recognition based on RepSViT.

### A. RepSViT Architecture

Following the principle of "ground training, on-orbit inference," the computationally intensive model training is performed on ground servers, while the on-orbit satellites focus on simple and fast online inference. Considering the stringent accuracy requirements of satellite on-orbit computations, the training structure of RepSViT should possess strong feature extraction capabilities. The overall architecture of RepSViT is illustrated in Fig. 1.

The overall architecture of RepSViT consists of spiking patch splitting (SPS), four stages of different scales, an ECFC Module, and global average pooling. SPS employs structural reparameterization techniques, where the training part follows the idea of "increasing network width to reduce network depth." It utilizes a three-branch network structure: the left branch is a $3 \times 3$ $D^2SC$, the middle branch is a $1 \times 1$ $D^2SC$, and the right branch is BatchNorm. The features extracted from the three branches are fused by addition and fed into the multistep leaky integrate-and-fire (LIF) layer and RepFNN for further feature extraction. The inference structure of SPS consists of two layers of $3 \times 3$ $D^2SC$-LIF and two layers of $1 \times 1$ $D^2SC$-LIF. The different branches of SPS extract features at different scales on various network widths, enhancing RepSViT's feature extraction capability. Stage 1, Stage 2, and Stage 3 have the same architecture, composed of BatchNorm, $3 \times 3$ $D^2SC$, RepFNN, and a residual connection that connects the input with the output of $D^2SC$. Their inference structure consists of a combination of one layer of $3 \times 3$ $D^2SC$, one layer of $3 \times 3$ $D^2SC$-LIF, and two layers of $1 \times 1$ $D^2SC$-LIF. Each layer of Patch Emb. follows a similar architecture, with two branches, namely, $7 \times 7$ $D^2SC$-BatchNorm and $3 \times 3$ $D^2SC$-BatchNorm, which are parallelly connected. They are then followed by multistep LIF, connected to a parallel structure of $1 \times 1$ $D^2SC$-BatchNorm and BatchNorm, and further concatenated with another multistep LIF. Patch Emb. utilizes large convolutional kernels and the dilated characteristics of $D^2SC$ to significantly improve the receptive field of the network and enhance feature extraction efficiency. Stage 1, Stage 2, and Stage 3 extract features at different depths of the network to capture features at different scales, thereby enhancing the model's feature extraction capability. Stage 4 consists of a conditional position encoder (CPE) [46], an improved self-attention mechanism (composed of BatchNorm and SGAM) along with a residual connection, and RepFNN.

### B. Dynamic Dilated Spiking Convolution

Dynamic convolution, due to its convolutional kernels being obtained by weighting multiple randomly initialized kernels, possesses enhanced feature extraction capability, leading to improved model performance [47]. Compared with standard convolution, dilated convolution can achieve a larger receptive field with fewer parameters, significantly enhancing feature extraction efficiency [48]. SNNs involve only additive computations and binary encoding characteristics, resulting in models developed based on SNNs being more computationally efficient and faster in operation compared

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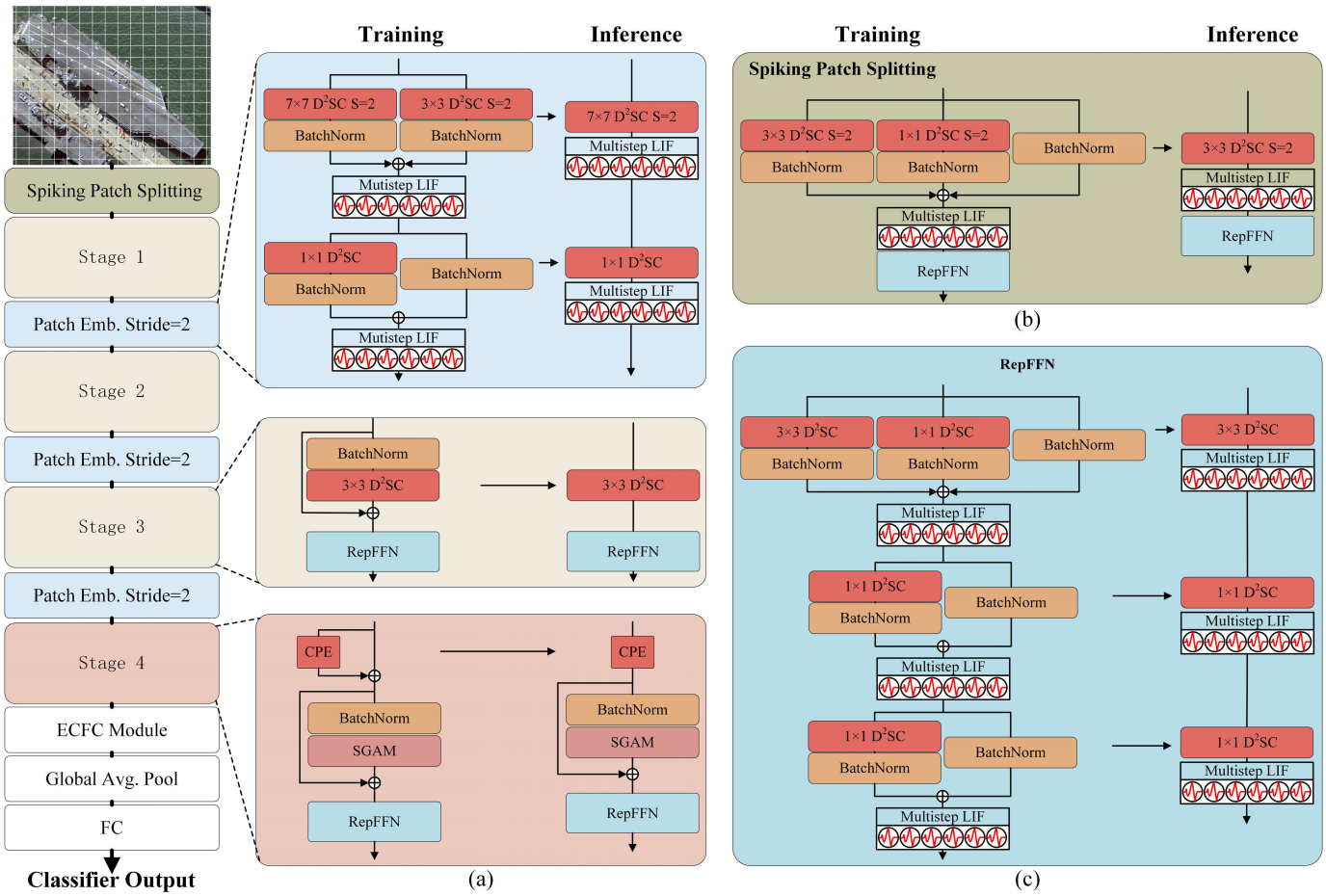



Fig. 1. Framework of the proposed RepSViT. The network structures during training and inference of (a) stage layer and Patch embedding layer, (b) spiking patch splitting, and (c) RepFFN.

with DNN models [39]. The advantages of dynamic convolution, dilated convolution, and SNNs align well with the requirements of "light, fast, and accurate" models for on-orbit satellite computing. Therefore, a natural idea arises to create a convolutional computing unit that combines the strengths of these three approaches, exhibiting strong feature extraction capabilities, high efficiency, low computational overhead, and fast inference speed. Building upon these insights, we have constructed D$^2$SC, as shown in Fig. 2.

D$^2$SC is composed of multistep LIF and dynamic dilated convolution in series. The dynamics of an LIF neuron can be formulated as follows:

$$H[t] = V[t-1] + \frac{1}{\tau}(X[t] - (V[t-1] - V_{\text{reset}})) \quad (1)$$

$$S[t] = \Theta(H[t] - V_{th}) \quad (2)$$

$$V[t] = H[t](1 - S[t]) + V_{\text{reset}}\, S[t] \quad (3)$$

where $\tau$ is the membrane time constant, and $X[t]$ is the input current at time step $t$. When the membrane potential $H[t]$ exceeds the firing threshold $V_{th}$, the spike neuron will trigger a spike $S[t]$. $\Theta(v)$ is the Heaviside step function, which equals 1 when $v \geq 0$ and 0, otherwise. $V[t]$ represents the membrane potential after the triggered event, which equals to $H[t]$ if no spike is generated and otherwise equals to the reset potential $V_{\text{reset}}$ . Multistep LIF first performs binary encoding on the input, enabling subsequent computations to be performed as spiking computations, thereby saving computational resources. Dynamic dilated convolution utilizes the single-kernel omnidimensional dynamic convolution (SODC) [17] to achieve powerful feature extraction while controlling its dilation rate to achieve efficient extraction of features with large receptive fields. The dilation rate of SODC in this article is manually tuned based on experimental results. Different from the general dynamic convolution that linearly aggregates multiple initialized convolution kernels, SODC generates convolution 4-D scale coefficients $\Pi$ $(\pi_{\text{out}}, \pi_{\text{in}}, \pi_h, \pi_w)$ of a randomly initialized convolution kernel $\mathrm{W} \in \mathbb{R}^{C_{\text{out}} \times (C_{\text{in}}/g) \times k \times k}$ through coefficient generation attention (CGA) learning input features, where $C_{\text{out}}, (C_{\text{in}}/g), k, k$ represent output channel, input_channel/groups, kennel height size and kernel width size, respectively. CGA uses the Cov1D operation to generate the scale coefficient $\pi_{\text{out}}, \pi_{\text{in}}$, which is lighter and faster and uses the full connection operation to generate the scale coefficient $\pi_h, \pi_w$. The specific derivation process of SODC operation is as follows.

$$\Pi(x) = \begin{cases} \text{Sigmoid}(\text{Conv1D}(\text{GAP}(x))), & x \in \mathbb{R}^{C_{\text{in}}} \\ \text{Sigmoid}(\text{FC}(\text{GAP}(x))), & x \in \mathbb{R}^{h \times w} \end{cases} \quad (4)$$

$$\text{SODC}(x) = \Pi(x) \times \mathrm{W} \times x + \pi_{\text{out}} \times b \quad (5)$$

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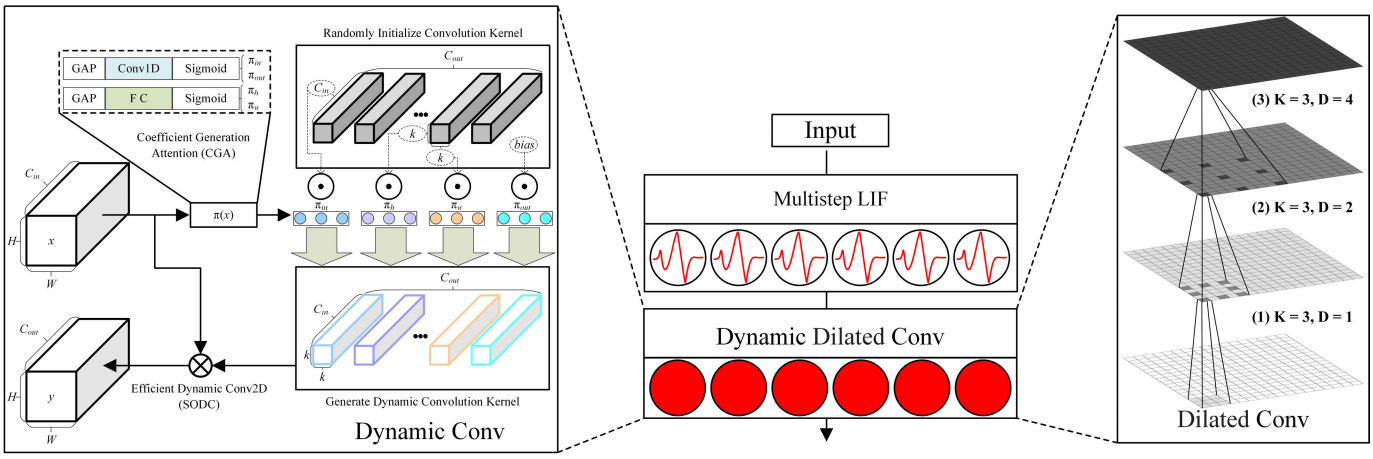



Fig. 2. Schematic of D$^2$SC.

where $x \in \mathbb{R}^{C_{\text{in}} \times h \times w}$ represents the input features, $W$ is a randomly initialized convolution kernel weight, $b$ is a randomly initialized bias term. The data flow of D$^2$SC could be formulated as follows:

$$O_l = D^2\text{SC}(\text{S N}_l(O_{l-1})) + O_{l-1} = S_l + O_{l-1} \tag{6}$$

$$O_{l+1} = D^2\text{SC}(\text{S N}_{l+1}(O_l)) + O_l = S_{l+1} + O_l \tag{7}$$

where $S_l$ denotes the D$^2$SC mapping learned as $S_l = D^2\text{SC}(\text{S N}_l(O_{l-1}))$; $\text{S N}_l(\cdot)$ represents the addition operation of the $l$ layer SNN; and $O_l$ is the output of the $l$ layer. $S_l + O_{l-1}$ belongs to the floating point addition operation, which is the same as the addition operation in the SNN layer. The floating point addition operation is the most essential operation of SNN. Obviously, the output $O_l$ is also a floating point and will pass through an SNN layer before participating in the next D$^2$SC computation. Therefore, the pure spike-form feature will be generated after the processing of the SNN layer and the computation of the D$^2$SC layer will become a pure floating point addition operation, following the spike-driven principle and reducing energy consumption vastly. Therefore, D$^2$SC combines the advantages of dynamic convolution for strong feature extraction, efficient feature extraction with large receptive fields using dilated convolution, and the computational efficiency and fast inference of SNNs, meeting the requirements of on-orbit computing models for satellites.

### C. Spiking Guided Attention Module

To further enhance the feature extraction capability and performance of RepSViT, we propose an improved and efficient attention mechanism called SGAM based on GAM [17]. As shown in Fig. 3, SGAM utilizes 1-D convolution to implement efficient channel attention (ECA) and employs D$^2$SC, which has strong and efficient feature extraction capability, to implement spatial attention. D$^2$SC enables binary encoding of the extracted features, making the application of the attention mechanism more computationally convenient. SGAM and GAM are both variants of convolutional block attention module (CBAM) [49], and their approaches to spatial attention differ from CBAM. CBAM uses 2-D standard convolutions to

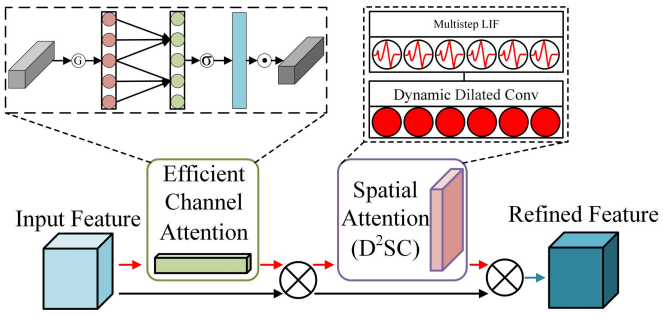



Fig. 3. Schematic of SGAM.

implement spatial attention. GAM utilizes SODC for spatial attention. In comparison with GAM, SGAM replaces SODC with D$^2$SC and introduces SNNs and dilated convolutions. Therefore, SGAM exhibits stronger and more efficient feature extraction capability.

SGAM uses a 1-D convolution operation to achieve ECA, allowing the model to focus more on the focused object feature channel. The required number of parameters $\text{Conv1D}_{\text{params}}$ is derived as follows:

$$\text{Conv1D}_{\text{params}} = k_{\text{size}} \times C_{\text{in}} \times C_{\text{out}} \qquad C_{\text{in}} \leqslant C_{\text{out}}. \tag{8}$$

And the derivation of the number of parameters $\text{SE}_{\text{params}}$ for the channel attention mechanism based on the fully connected layer is as follows:

$$\text{SE}_{\text{params}} = \frac{2}{r} \times C_{\text{out}}^2 \qquad 0 < r < 1. \tag{9}$$

Divide (8) by (9) to get

$$\text{Compression}_{\text{ratio}} = \frac{\text{Conv1D}_{\text{params}}}{\text{SE}_{\text{params}}} = \frac{r}{2} \times \frac{C_{\text{in}}}{C_{\text{out}}} < 1 \tag{10}$$

where $C_{\text{in}}$ is the input feature channel, $k_{\text{size}} = 1$ is the kernel size of the 1-D convolution (when bias terms are used in Cov1D and SE [50], the introduced parameters and the computational costs are typically negligible). $r$ is the reduction rate, and $C_{\text{out}}$ is the number of output channels.

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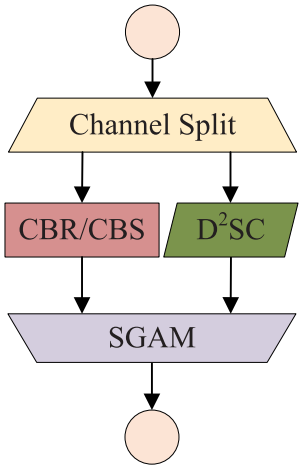



Fig. 4. Schematic of ECFC.

Equation (8) shows that the number of parameters of the 1-D convolution is a first-order function of $C_{\text{out}}$, while (9) shows that the number of parameters of the fully connected layer is a second-order function of $C_{\text{out}}$. It can be seen from (10) that the $\text{Compression}_{\text{ratio}}$ is less than 1, i.e., as the number of output channels increases, the number of parameters of the fully connected layer channel attention will be much larger than that of the 1-D convolution channel attention. In addition, the 1-D convolution directly extracts the data features, unlike the fully connected layer which requires feature scaling using the reduction rate $r$ and then indirectly extracts the features, avoiding feature dimensionality reduction and efficiently extracting the features.

### *D. Efficient Coupled Fine-Coarse-Grained Feature Extraction*

To enhance the efficient extraction of multiscale features in RepSViT, we have designed the ECFC module, which is an efficient module for coupling coarse-grained and fine-grained feature extraction in RepSViT. As shown in Fig. 4, the input of the ECFC module is split into two parallel branches in a 1:1 ratio. The left branch consists of CBR/CBS, where CBR represents a $1 \times 1$ standard convolution, BatchNorm, and ReLU. (In CBS, the "S" denotes the SiLU activation function, which is more suitable for deeper layers in the network.) CBR performs better than CBS in shallow feature extraction and CBS performs better in deep feature extraction. In this article, ECFC is operated in the deep layer of the network so it uses CBS operation. The left branch extracts fine-grained features. The right branch employs a $3 \times 3$ $D^2SC$ operation to extract coarse-grained features. Subsequently, the coarse-grained and fine-grained features extracted from the left and right branches are fused using SGAM to obtain the output of the ECFC module, which represents the coupled coarse-grained and fine-grained features.

### *E. Reparameterized Feed-Forward Network*

To enhance the feature extraction capability during the training phase and improve the inference speed during the inference phase of RepSViT, we have designed RepFFN, which is based on the structural reparameterization technique. During the training phase, it employs a complex branched structure to increase the feature extraction capability of RepSViT. During the inference phase, it adopts a streamlined structure with a single pathway to reduce memory access and thereby improve the inference speed of RepSViT.

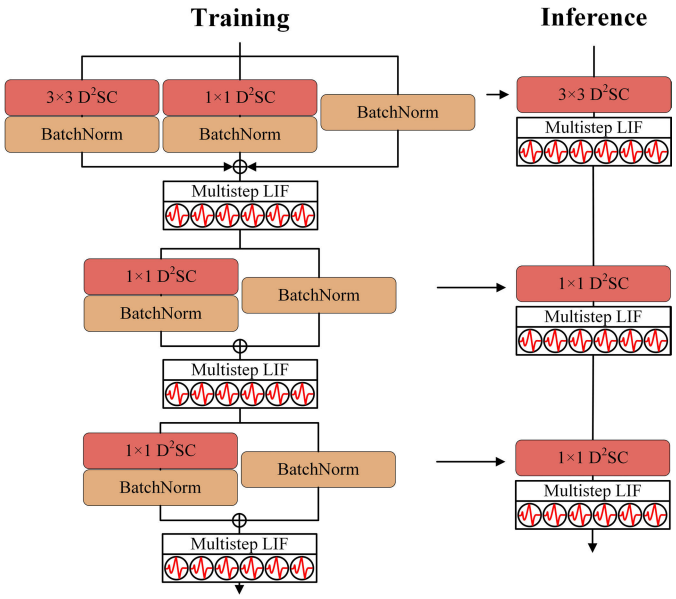



Fig. 5. Schematic of RepFFN.

As illustrated in Fig. 5, the structure of RepFFN during the training phase consists of a three-branch structure similar to SPS and two two-branch parallel structures similar to Patch Emb. These structures are then interconnected using multistep LIF, with one three-branch structure followed by two two-branch structures. The three-branch structure is responsible for extracting features at different scales, while the two-branch parallel structure refines the extracted features at different scales. The multistep LIF component plays a role in the binary encoding of the features, thereby accelerating the computational speed.

### *F. Satellite On-Orbit Object Recognition Based on RepSViT*

The RepSViT proposed in this article is specially developed and designed for satellite on-orbit object recognition. The implementation process of satellite on-orbit object recognition technology is shown in Fig. 6, which is divided into four stages: 1) training on the ground server. In the first stage, we construct the historical RSI dataset and use it to train and tune RepSViT on the ground server, then obtain the trained model to prepare for the second stage. 2) Updating the model on the satellite. In the second stage, we regularly update the latest model trained on the ground to the satellite so that the on-board model can detect new objects as early as possible. First, encapsulate the latest optimal recognition model in the ground system and send it to the satellite service through the satellite-ground uplink channel. After the satellite service system recognizes the data as the intelligent on-orbit processor's on-orbit data according to the agreed identification,

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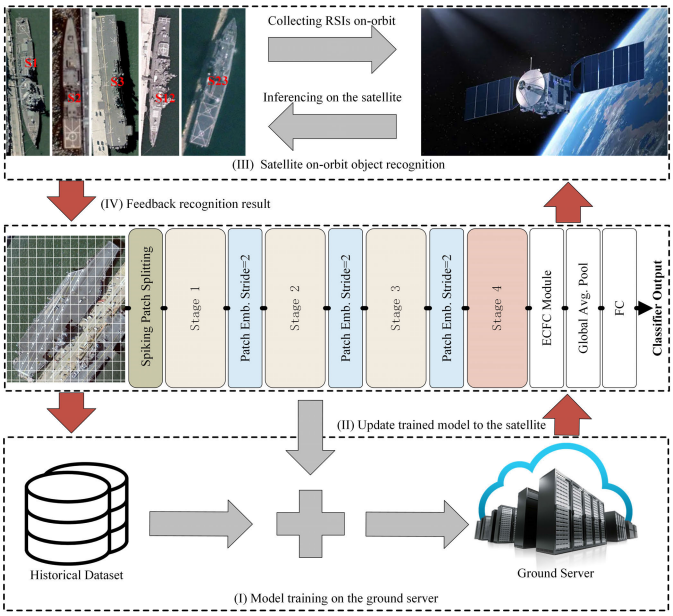



Fig. 6. Satellite on-orbit object recognition based on RepSViT.

the data are transmitted to the control unit of the intelligent on-orbit processing controller, and the control unit judges the validity of the received data, if it is valid, it will be transmitted to the computing unit, otherwise, it will be discarded and fed back by telemetry. Finally, the computing unit centrally caches the received data and performs model loading of the received data and on-orbit inferencing according to the control instruction after receiving is completed. 3) Inferencing on the satellite. In the third stage, we use the optimal model obtained in stage two and deploy it on the satellite board. The RSIs obtained on-orbit will be directly used for object detection without being transmitted to the ground. 4) Feedback detected results to users and service stations on the ground. In this stage, the satellite will send the detected object information, including object category, object coordinates, and object image slices to the users and the ground service center in real time.

This method of satellite-on-orbit object recognition has changed the traditional RSIs satellite-to-ground transmission and ground processing application mode, greatly improving the timeliness of satellite response.

## IV. Experimental Settings

In this section, the proposed RepSViT is verified on the ground server, and the satellite is imitated by TX2, the results of the experiment, including the ablation study and some comparison experiments of state-of-the-art lightweight methods, will be presented and analyzed.

### A. Data Description

Fine-Grained Ship Collection-23 (FGSC-23) [51] is a high-resolution optical RSI dataset for fine-grained ship object recognition. It consists of 23 ship categories with a total of 4052 ship instances. The data is sourced from high-resolution Google Earth and GF-2 satellite images. To meet the requirements of different models regarding input image size, we uniformly resized the images to 224 × 224.

RSI Object Recognition-9 (RSIOR-9) dataset is a subdataset selected from the NWPU-RESISC45 dataset [52] for complex background object recognition and contains a total of nine object categories, airplane, airport, bridge, freeway, harbor, island, railway, railway station, and ship. Each class includes 700 images with a size of 256 × 256 pixels in the red–green–blue (RGB) color space.

We divided the data into training, validation, and testing sets in a ratio of 3:1:1. To fully showcase the performance of different models, we applied geometric transformations as data augmentation techniques to the training dataset. These transformations include random horizontal flipping, random rotation up to 30°, random size crop and resize, random affine transformation up to 30°, and scaling by a factor of 1/0.877. However, the testing set remains in its original state without data augmentation to reflect the performance of models under real data conditions.

### B. Performance Metric

In this article, the fine-grained recognition task is a multiclass classification problem. To evaluate the model performance with unbalanced data, we adopt macroaverage F1-score ( Macro $F1$), macroaverage Precision (Macro $P$), and macroaverage Recall (Macro $R$) as performance metrics. Macro $F1$ is a comprehensive metric to evaluate the overall performance in all classes. The f1-score is a comprehensive performance metric for each subcategory. Accuracy is the overall accuracy of all categories. The specific calculation formula is as follows:

$$\text{Precision(P)} = \frac{t_p}{t_p + f_p} \tag{11}$$

$$\text{Recall(R)} = \frac{t_p}{t_p + f_n} \tag{12}$$

$$f1 - \text{score} = \frac{2 \times P \times R}{P + R} \tag{13}$$

$$\text{Accuracy} = \frac{\sum_{i=1}^{x} t_{\text{pi}} + t_{\text{ni}}}{\sum_{i=1}^{x} t_{\text{pi}} + t_{\text{ni}} + f_{\text{pi}} + f_{\text{ni}}} \tag{14}$$

$$\text{Macro } P = \frac{1}{x} \sum_{i=1}^{x} P_i \tag{15}$$

$$\text{Macro } R = \frac{1}{x} \sum_{i=1}^{x} R_i \tag{16}$$

$$\text{Macro } F1 = \frac{2 \times \text{ Macro } P \times \text{ Macro } R}{\text{Macro } P + \text{ Macro } R} \tag{17}$$

where $t_p$, $t_n$, $f_p$, and $f_n$ refer to true positives, true negatives, false positives, and false negatives in class $i$, respectively. Herein, $x$ is the number of classes to be classified.

We calculate the number of the synaptic operations of spike (SOP) to evaluate the computational cost of spiking layers, with the following formula:

$$\text{SOP}^l = \text{fr} \times T \times \text{FLOPs}^l \tag{18}$$

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TABLE I
COST AND GAIN OF EACH IMPROVEMENT STEP

| Model | D$^2$SC | SGAM | ECFC | RepFFN | Latency(ms) | FLOPs(G) | Params(M) | Accuracy | Macro P | Macro R | Macro F1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Baseline | - | - | - | - | 1.33 | 0.71 | 3.62 | 0.83 | 0.84 | 0.83 | 0.83 |
| D$^2$SCViT | ✓ | - | - | - | 0.72(↓45%) | 0.45(↓36%) | 3.87(↑6.9%) | 0.91(↑8%) | 0.92 | 0.91 | 0.91 |
| SGAMViT | ✓ | ✓ | - | - | 0.85(↑18%) | 0.46(↑2.2%) | 3.72(↓3.8%) | 0.93(↑2%) | 0.93 | 0.93 | 0.93 |
| ECFCViT | ✓ | ✓ | ✓ | - | 1.07(↑25%) | 0.58(↑26%) | 3.75(↑0.8%) | 0.94(↑1%) | 0.94 | 0.94 | 0.94 |
| RepSViT | ✓ | ✓ | ✓ | ✓ | 1.02(↓4.6%) | 0.60(↑3.4%) | 3.77(↑0.5%) | **0.95**(↑1%) | 0.95 | 0.95 | 0.95 |

where $l$ is a spiking layer in RepSViT, fr is the firing rate of the layer and $T$ is the simulation time step of spike neuron. FLOPs$^l$ refers to floating point operations of layer $l$, which is the number of multiply-and-accumulate (MAC) operations. And SOP$^l$ is the number of spike-based accumulate (AC) operations. When calculating the computational cost in GFLOPs for RepSViT, we accounted for all multiplication operations, including operations involving zero multiplication; the computational cost of spiking layers is determined using the SOP from (18).

### C. Training Details

All experimental model training is carried out on the ground server. The server configuration is as follows: 8 × 10-core Intel Xeon Gold 6148 @2.40-GHz CPU, 8× NVIDIA GeForce RTX 3090 GPUs, and an operating system of 64-bit CentOS Linux release 7.9.2009, the programming language is Python-3.7.13, and the deep learning framework is PyTorch-1.12.1. The models for conducting experiments are implemented based on SpikingJelly and Timm.

The hyperparameters of the model are obtained by a grid search with the following range: batch size: [16,32,64,128], epochs: [100,200,300,400,500], optimizer: [adamw], and we establish a fivefold cross-validation. The model hyperparameters are set as follows: surrogated gradients are used to train spiking neurons, the time step is 8, the optimizer is adamw, the initial learning rate is 5e-4, the attenuation is 0.1 at 200 and 300 epoch, the batch_size is 64, the epoch is 400, the loss function is cross-entropy, and the random seed is 42.

### D. Ablation Study

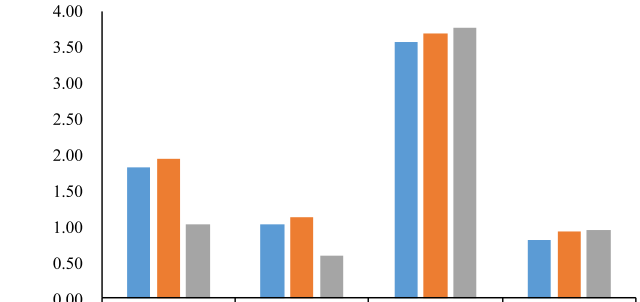
To prove the roles of each module or computing unit designed in RepSViT, we performed ablation experiments on D$^2$SC, SGAM, ECFC, and RepFFN, respectively, and used FastViT [18], which is similar to the RepSViT architecture, as the baseline.

From the experimental results in Table I, we observed the impact of each module designed in RepSViT. D$^2$SCViT, which employed only D$^2$SC, achieved an 84.72% faster inference speed than the baseline, with 63.38% of the computational costs of the baseline. Additionally, D$^2$SCViT showed an 8% accuracy improvement compared with the baseline, with a parameter increase of only 0.25 M. These findings indicate that D$^2$SC significantly enhances model inference speed, reduces the computational costs, and improves recognition accuracy with a minimal increase in parameters. SGAMViT, incorporating both D$^2$SC and SGAM, showed only marginal changes in inference speed, computational costs, and parameters compared with D$^2$SCViT. However, it demonstrated a 2% accuracy increase, indicating that SGAM can better extract object features and consequently enhance model performance. ECFCViT, which utilized D$^2$SC, SGAM, and ECFC, exhibited slight increases in inference latency, computational costs, and parameters compared with SGAMViT. Meanwhile, it achieved a 1% accuracy improvement, demonstrating that ECFC can better extract coarse-fine-grained features to enrich the diversity of object features and improve model performance. Finally, RepSViT, incorporating all the modules designed in this article, achieved lower inference latency compared with ECFCViT, with only slight increases in computational costs and parameters, along with a 1% accuracy improvement. These results indicate that RepFFN, utilizing reparameterization techniques, enhances model inference speed and feature extraction capability, thereby improving model performance.

*1) Study of the Dynamic Dilated Spiking Convolution:* To demonstrate the effectiveness and superiority of D$^2$SC proposed in this article, we conducted experiments by, respectively, replacing D$^2$SC with standard convolutions (SCViT) and SODC (SODCViT) while keeping all other settings identical. From the experimental results shown in Fig. 7, it can be observed that the inference latency of RepSViT using D$^2$SC is 0.81 ms lower than SCViT and 0.92 ms lower than SODCViT. In terms of computational costs, RepSViT consumes 0.42 GFLOPs less than SCViT and 0.53 GFLOPs less than SODCViT. RepSViT achieves a recognition accuracy of 95%, which is 14% higher than SCViT and 3% higher than SODCViT. These findings indicate that the proposed D$^2$SC is capable of reducing inference latency and computational costs, while also improving model accuracy. These findings indicate that the proposed D$^2$SC not only reduces inference latency and computational costs but also improves model accuracy.

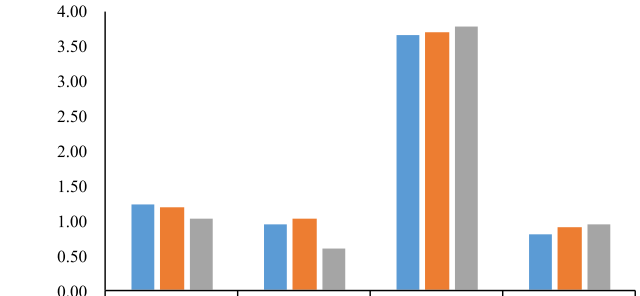
*2) Study of Spiking Guided Attention Module:* To demonstrate the effectiveness and superiority of the proposed SGAM, we conducted experiments where SGAM was, respectively, replaced by CBAM (CBAMViT) and GAM (GAMViT) while keeping all other settings identical. The experimental results, as depicted in Fig. 8, reveal that using SGAM in RepSViT leads to a reduction in inference latency by 0.21 ms compared with CBAMViT and 0.16 ms compared with GAMViT. Additionally, RepSViT with SGAM exhibits a decrease in computational costs by 0.35 GFLOPs compared with CBAMViT and by 0.42 GFLOPs compared with GAMViT. Furthermore, RepSViT with SGAM achieves higher

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| | Latency(ms) | FLOPs(G) | Params(M) | Accuracy |
|---|---|---|---|---|
| SCViT | 1.83 | 1.02 | 3.57 | 0.81 |
| SODCViT | 1.94 | 1.13 | 3.69 | 0.92 |
| RepSViT | 1.02 | 0.60 | 3.77 | 0.95 |

Fig. 7. Results of ablation experiments on $D^2SC$.




| | Latency(ms) | FLOPs(G) | Params(M) | Accuracy |
|---|---|---|---|---|
| CBAMViT | 1.23 | 0.95 | 3.66 | 0.80 |
| GAMViT | 1.18 | 1.02 | 3.70 | 0.90 |
| RepSViT | 1.02 | 0.60 | 3.77 | 0.95 |

Fig. 8. Results of ablation experiments on SGAM.

accuracy, surpassing CBAMViT by 15%, and outperforming GAMViT by 5%. From these observations, it can be inferred that the proposed SGAM contributes to a slight reduction in inference latency and computational costs while significantly improving model accuracy.

*3) Study of Efficient Coupled Fine-Coarse-Grained Block:* To demonstrate the effectiveness and superiority of the proposed ECFC in this study, we conducted experiments by replacing ECFC with CFC (CFCViT) while keeping all other settings identical. The structural comparison diagram of CFC and ECFC is shown in Fig. 9. CFC employs depthwise separable hybrid dilated convolution (DSHDC) [3] for coarse-grained feature extraction and utilizes GAM for coarse-to-fine feature fusion; whereas ECFC employs $D^2SC$ for coarse-grained feature extraction and uses SGAM for coarse-to-fine feature fusion. The experimental results are shown in Fig. 10, where we observed that the inference latency of RepSViT using ECFC is 0.26 ms lower than CFCViT. Additionally, RepSViT consumes 0.27 GFLOPs less in terms of computational costs compared with CFCViT. Furthermore, RepSViT achieves a 9% higher accuracy than CFCViT. These findings indicate that the proposed ECFC contributes to slight reductions in inference latency and computational costs while significantly improving model accuracy.

In the ECFC module, two parallel branches are used for coarse and fine-grained feature extraction, with a ratio set at 1:1. The 1:1 ratio we employed is inspired by the research of

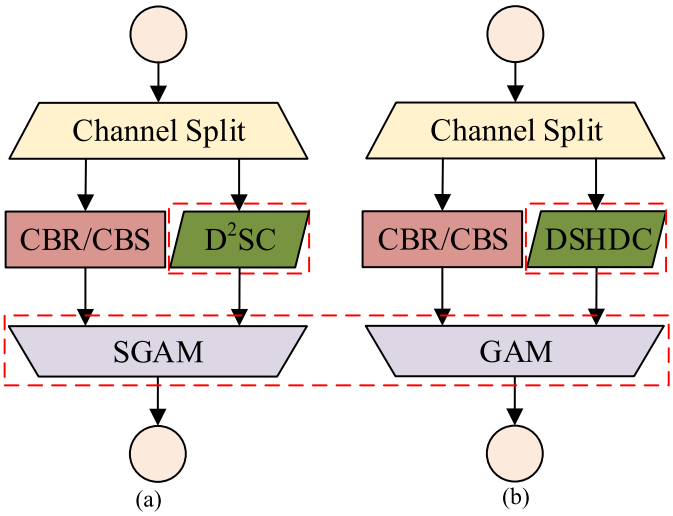
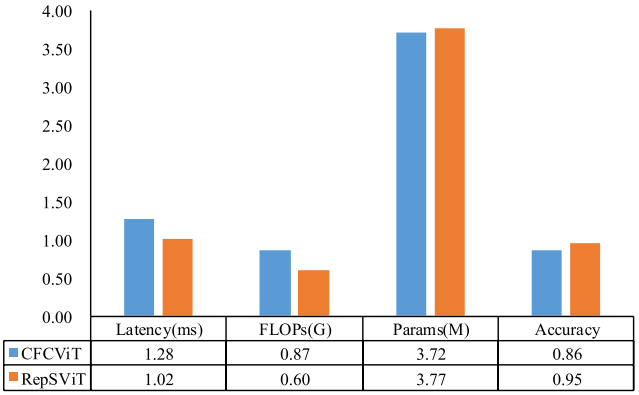



Fig. 9. Structural comparison diagram of CFC and ECFC. (a) Structure of ECFC. (b) Structure of CFC.



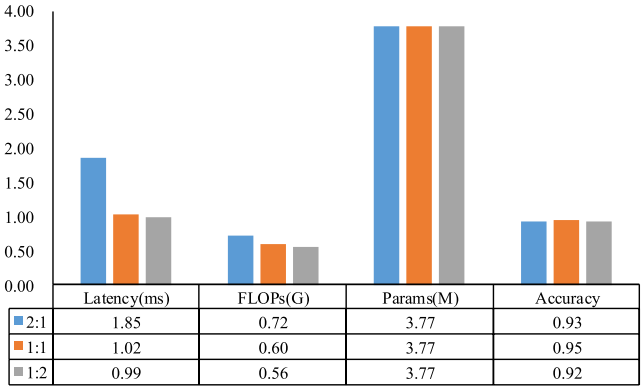

| | Latency(ms) | FLOPs(G) | Params(M) | Accuracy |
|---|---|---|---|---|
| CFCViT | 1.28 | 0.87 | 3.72 | 0.86 |
| RepSViT | 1.02 | 0.60 | 3.77 | 0.95 |

Fig. 10. Results of ablation experiments on ECFC.

ShuffleNetv2 [53]. To validate the rationale and superiority of the 1:1 ratio, we conducted experiments using three different ratios: 2:1, 1:1, and 1:2. The experimental results are illustrated in Fig. 11. From Fig. 11, it is evident that the ECFC with a ratio of 2:1 exhibits a slower inference speed compared with the 1:1 ratio counterpart, and the former incurs higher computational costs than the latter. This observation suggests that the left branch of ECFC, engaged in fine-grained feature extraction through $1 \times 1$ convolution, imposes a substantial computational burden, thereby diminishing the inference speed. Consequently, the ECFC with a 1:2 ratio demonstrates a faster inference speed than the 1:1 ratio configuration, with lower computational costs. However, regardless of whether the ECFC ratio is 2:1 or 1:2, both configurations exhibit lower recognition accuracy than the 1:1 ratio ECFC. This is attributed to the 2:1 ratio, causing the model to overly focus on intricate details of the object features at the expense of global features. Conversely, the 1:2 ratio results in the model emphasizing global features while neglecting the finer details of the object.

To further substantiate the advantages of ECFC in multiscale feature extraction, we perform an ablation experiment by modifying the ECFC structure. Specifically, we eliminate the

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| | Latency(ms) | FLOPs(G) | Params(M) | Accuracy |
|---|---|---|---|---|
| 2:1 | 1.85 | 0.72 | 3.77 | 0.93 |
| 1:1 | 1.02 | 0.60 | 3.77 | 0.95 |
| 1:2 | 0.99 | 0.56 | 3.77 | 0.92 |

Fig. 11. Results of ablation experiments on three different ratios of ECFC.

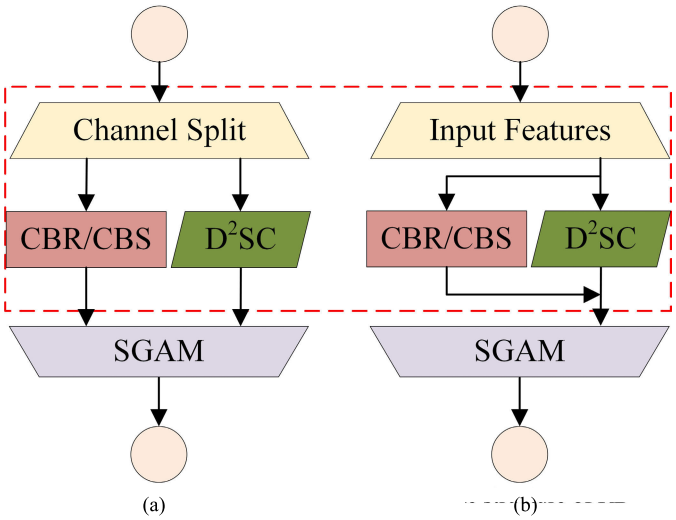
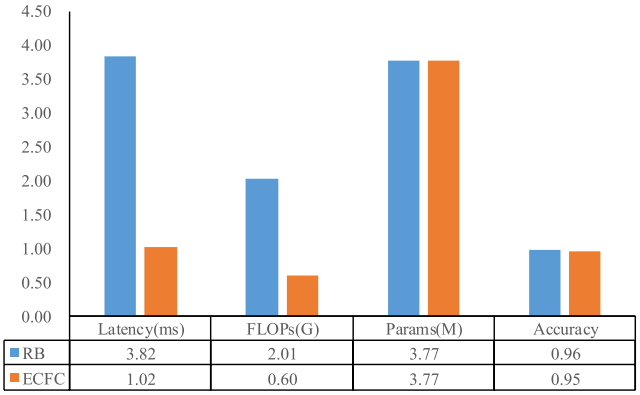



Fig. 12. Comparison of ECFC and RB structures. (a) Structure of ECFC. (b) Structure of RB.

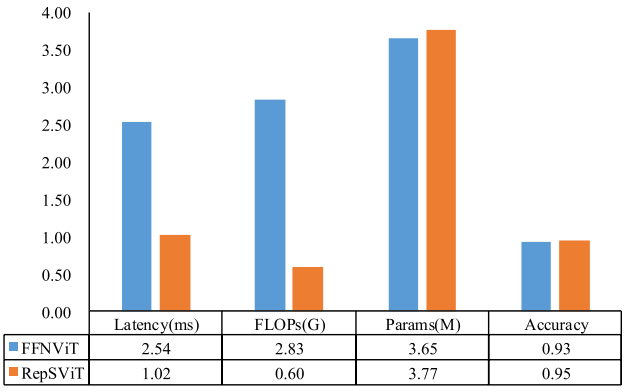
channel split operation from the ECFC structure, and the data flow is directly input into D2SC and CBR/CBS. The outputs of these two paths are then summed and fed into SGAM, resembling the ResNet block (RB) structure illustrated in Fig. 12. The experimental results, as illustrated in Fig. 13, reveal that the adoption of the RB structure in RepSViT leads to a 3.745-fold increase in latency, a 3.35-fold rise in computational cost, a 1% improvement in accuracy, with parameter quantity exhibiting minimal change. Consequently, the RepSViT utilizing the ECFC structure, in comparison with its RB structure counterpart, demonstrates significant advantages in conserving computational resources and enhancing inference speed, thereby validating its efficacy.

*4) Study of Reparameterized Feed-Forward Network:* To demonstrate the effectiveness and superiority of the proposed RepFFN, we conducted experiments by replacing RepFFN with FFN (FFNViT) while keeping all other settings identical. The experimental results are shown in Fig. 14. From Fig. 14, we observe that using RepFFN in RepSViT reduces the inference latency by 1.52 ms compared with FFNViT. Additionally, RepSViT consumes 2.23 GFLOPs less in terms of computational resources than FFNViT. Moreover, RepSViT achieves a 2% higher accuracy than FFNViT. These findings indicate that




| | Latency(ms) | FLOPs(G) | Params(M) | Accuracy |
|---|---|---|---|---|
| RB | 3.82 | 2.01 | 3.77 | 0.96 |
| ECFC | 1.02 | 0.60 | 3.77 | 0.95 |

Fig. 13. Results of ablation experiments on multiscale features extraction of ECFC.




| | Latency(ms) | FLOPs(G) | Params(M) | Accuracy |
|---|---|---|---|---|
| FFNViT | 2.54 | 2.83 | 3.65 | 0.93 |
| RepSViT | 1.02 | 0.60 | 3.77 | 0.95 |

Fig. 14. Results of ablation experiments on RepFFN.

the proposed RepFFN significantly reduces inference latency and computational costs.

*5) Study of the Number of Time Steps:* To explore the tradeoff between accuracy and latency with changes in the number of time steps, experiments are conducted with time steps of 4, 8, 16, 32, 64, and 128, as illustrated in Fig. 15. We observe that with a time step of 4, the accuracy is only 77%, although the latency is relatively low; the accuracy for time steps 16 and 8 is both 95%, but the latency for time step 16 is almost twice that of time step 8; although the accuracy for time steps 32, 64, and 128 is slightly higher than that of time step 8, their latency increases significantly. In summary, the experimental results with a time step of 8 balance accuracy and latency.

### *E. Comparison With State-of-the-Art Methods*

To further demonstrate the effectiveness and superiority of the proposed RepSViT, we conduct experiments based on the NVIDIA Jetson TX2 satellite board on two datasets, FGSC-23 and RSIOR-9, and analyze them in comparison with the state-of-the-art lightweight ViT methods, SNN methods, and SOCNet method, which are applied to RSI recognition.

*1) Validation on the FGSC-23 Dataset:* Table II presents the recognition performance of all methods for each class of ship, providing a detailed overview of the recognition results of various methods. Table II and Figs. 16 and 17 further confirm the effectiveness and superiority of RepSViT.

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TABLE II
DETAILED EXPERIMENTAL RESULTS OF ALL MODELS ON THE FGSC-23 DATASET

| Model | Category | S0 | S1 | S2 | S3 | S4 | S5 | S6 | S7 | S8 | S9 | S10 | S11 | S12 | S13 | S14 | S15 | S16 | S17 | S18 | S19 | S20 | S21 | S22 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FastViT | Precision | 0.84 | 0.83 | 0.79 | 0.82 | 0.89 | 0.78 | 0.84 | 0.86 | 0.80 | 0.80 | 0.81 | 0.90 | 0.80 | 0.91 | 0.82 | 0.83 | 0.83 | 0.86 | 0.82 | 0.81 | 0.86 | 0.86 | 0.87 |
| | Recall | **0.98** | **0.96** | 0.79 | 0.95 | 0.88 | 0.69 | 0.83 | 0.87 | 0.86 | 0.86 | 0.96 | 0.95 | 0.74 | 0.69 | 0.81 | 0.69 | 0.88 | 0.78 | 0.76 | 0.74 | 0.94 | 0.84 | 0.75 |
| | F1-Score | 0.91 | 0.89 | 0.79 | 0.88 | 0.88 | 0.73 | 0.83 | 0.87 | 0.83 | 0.83 | 0.88 | 0.92 | 0.77 | 0.78 | 0.81 | 0.75 | 0.85 | 0.82 | 0.79 | 0.77 | 0.90 | 0.85 | 0.81 |
| lightViT | Precision | 0.78 | 0.67 | 0.78 | 0.81 | 0.71 | 0.60 | 0.75 | 0.81 | 0.72 | 0.80 | 0.63 | 0.68 | 0.69 | 0.77 | 0.75 | 0.80 | 0.78 | 0.72 | 0.63 | 0.79 | 0.77 | 0.70 | 0.82 |
| | Recall | 0.62 | 0.62 | **0.97** | 0.96 | 0.79 | 0.50 | 0.72 | 0.75 | 0.53 | **1.00** | 0.57 | 0.89 | 0.60 | **0.92** | 0.80 | 0.55 | 0.90 | 0.73 | 0.50 | 0.89 | 0.82 | 0.63 | 0.77 |
| | F1-Score | 0.69 | 0.65 | 0.86 | 0.88 | 0.75 | 0.55 | 0.73 | 0.78 | 0.61 | 0.89 | 0.60 | 0.77 | 0.64 | 0.84 | 0.77 | 0.65 | 0.83 | 0.72 | 0.56 | 0.84 | 0.79 | 0.66 | 0.79 |
| EdgeViT | Precision | 0.82 | 0.85 | 0.86 | 0.79 | 0.88 | 0.81 | 0.79 | 0.87 | 0.84 | 0.89 | 0.84 | 0.84 | 0.80 | 0.84 | 0.78 | 0.91 | 0.88 | 0.81 | 0.82 | 0.81 | 0.81 | 0.82 | 0.88 |
| | Recall | 0.73 | 0.93 | 0.96 | 0.88 | **1.00** | 0.88 | 0.79 | 0.90 | 0.81 | **1.00** | 0.94 | 0.66 | 0.64 | 0.63 | 0.82 | **1.00** | 0.66 | 0.97 | 0.76 | 0.96 | 0.76 | 0.59 | 0.96 |
| | F1-Score | 0.77 | 0.89 | 0.91 | 0.83 | 0.94 | 0.85 | 0.79 | 0.88 | 0.82 | **0.94** | 0.89 | 0.74 | 0.71 | 0.72 | 0.80 | 0.95 | 0.75 | 0.88 | 0.79 | 0.88 | 0.78 | 0.69 | 0.92 |
| Spikformer | Precision | 0.70 | 0.67 | 0.74 | 0.71 | 0.69 | 0.69 | 0.71 | 0.80 | 0.73 | 0.81 | 0.68 | 0.73 | 0.70 | 0.72 | 0.75 | 0.70 | 0.73 | 0.65 | 0.78 | 0.78 | 0.83 | 0.73 | 0.74 |
| | Recall | 0.70 | 0.54 | 0.86 | 0.53 | 0.73 | 0.74 | 0.67 | **0.97** | 0.69 | 0.93 | 0.69 | 0.53 | 0.66 | **0.92** | 0.58 | 0.76 | 0.81 | 0.57 | 0.56 | 0.95 | 0.73 | 0.85 | 0.81 |
| | F1-Score | 0.70 | 0.60 | 0.79 | 0.61 | 0.71 | 0.71 | 0.69 | 0.87 | 0.71 | 0.87 | 0.68 | 0.61 | 0.68 | 0.81 | 0.66 | 0.73 | 0.77 | 0.61 | 0.65 | 0.86 | 0.78 | 0.78 | 0.77 |
| Spikingformer | Precision | 0.82 | 0.86 | 0.80 | 0.82 | 0.84 | 0.80 | 0.81 | 0.81 | 0.81 | 0.76 | 0.82 | 0.75 | 0.76 | 0.80 | 0.81 | 0.86 | 0.78 | 0.81 | 0.75 | 0.86 | 0.78 | 0.85 | 0.84 |
| | Recall | 0.93 | 0.93 | 0.86 | 0.94 | 0.83 | 0.80 | 0.75 | 0.80 | 0.76 | 0.65 | **0.98** | 0.80 | 0.90 | 0.77 | 0.66 | 0.65 | 0.81 | 0.69 | 0.79 | 0.89 | 0.68 | **0.96** | 0.77 |
| | F1-Score | 0.87 | 0.89 | 0.83 | 0.87 | 0.83 | 0.80 | 0.78 | 0.80 | 0.78 | 0.70 | 0.89 | 0.78 | 0.82 | 0.79 | 0.73 | 0.74 | 0.79 | 0.75 | 0.77 | 0.88 | 0.73 | 0.90 | 0.80 |
| SOCNet | Precision | 0.88 | 0.92 | **0.91** | 0.88 | 0.91 | 0.91 | 0.94 | 0.91 | **0.95** | 0.95 | 0.92 | 0.92 | 0.90 | 0.85 | 0.89 | 0.90 | 0.91 | 0.88 | 0.91 | 0.87 | 0.91 | 0.91 | 0.90 |
| | Recall | 0.89 | **0.96** | 0.90 | **0.99** | 0.86 | 0.86 | 0.83 | 0.86 | **0.97** | 0.88 | 0.89 | 0.95 | **1.00** | 0.88 | 0.90 | 0.94 | 0.87 | **1.00** | 0.87 | 0.95 | 0.82 | 0.87 | 0.87 |
| | F1-Score | 0.89 | 0.94 | 0.90 | 0.93 | 0.88 | 0.88 | 0.88 | 0.89 | **0.96** | 0.91 | 0.90 | 0.94 | **0.95** | 0.87 | 0.90 | 0.92 | 0.89 | **0.94** | 0.89 | 0.91 | 0.86 | 0.89 | 0.88 |
| RepSViT | Precision | **0.98** | **0.96** | **0.91** | **0.93** | **0.92** | **0.93** | **0.96** | **0.97** | 0.93 | **0.97** | **0.98** | **0.97** | **0.95** | **0.92** | **0.98** | **0.95** | **0.92** | **0.94** | **0.97** | **0.97** | **0.96** | **0.96** | **0.96** |
| | Recall | 0.90 | 0.93 | 0.96 | 0.96 | 0.98 | **0.98** | **0.99** | 0.90 | 0.91 | 0.91 | 0.90 | **0.97** | 0.92 | 0.90 | **0.98** | 0.96 | **0.99** | 0.94 | **0.99** | **1.00** | **0.96** | 0.95 | **1.00** |
| | F1-Score | **0.94** | **0.94** | **0.94** | **0.95** | **0.95** | **0.96** | **0.98** | **0.93** | 0.92 | **0.94** | **0.94** | **0.97** | 0.93 | **0.91** | **0.98** | **0.96** | **0.95** | **0.94** | **0.98** | **0.99** | **0.96** | **0.95** | **0.98** |

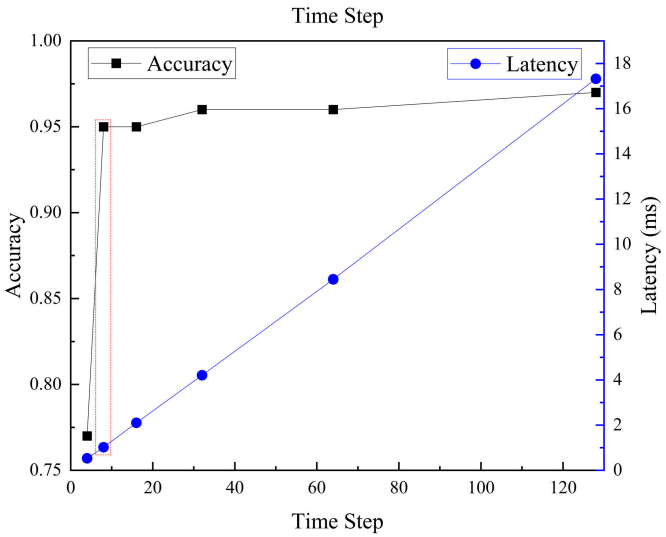



Fig. 15. Results of ablation experiments on time steps.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE FGSC-23

| Model | Latency(ms) | FLOPs(G) | Params(M) | Accuracy | Macro P | Macro R | Macro F1 |
|---|---|---|---|---|---|---|---|
| FastViT-T8 [18] | 8.51 | 0.71 | 3.62 | 0.83 | 0.84 | 0.83 | 0.83 |
| LightViT [19] | 15.82 | 0.73 | 9.40 | 0.74 | 0.74 | 0.74 | 0.73 |
| EdgeViT [20] | 16.28 | 0.56 | 4.10 | 0.84 | 0.84 | 0.84 | 0.83 |
| Spikformer [54] | 15.32 | 6.82 | 16.81 | 0.81 | 0.81 | 0.81 | 0.81 |
| SpikingFormer [22] | 7.47 | 3.88 | 16.81 | 0.73 | 0.73 | 0.73 | 0.72 |
| SOCNet [3] | **2.45** | **0.05** | **0.21** | 0.90 | 0.91 | 0.90 | 0.90 |
| RepSViT | 8.33 | 0.60 | 3.77 | **0.95** | **0.95** | **0.95** | **0.95** |

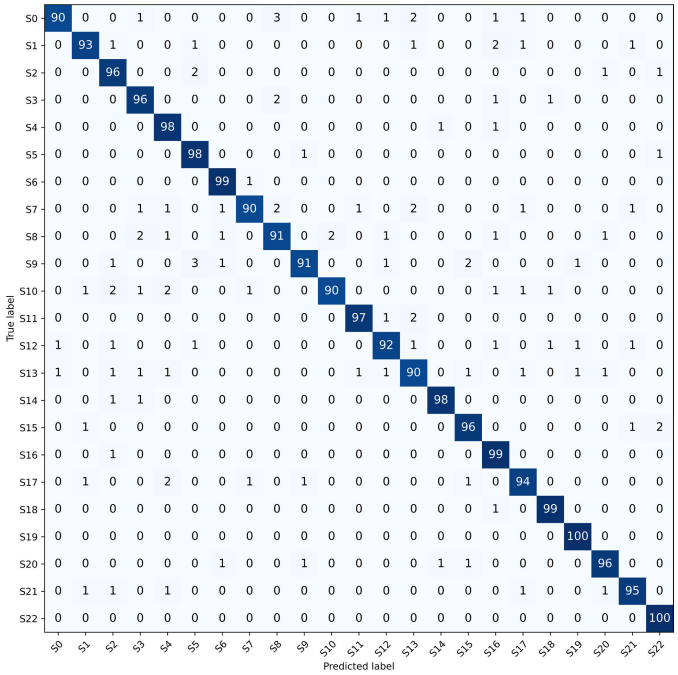



Fig. 16. Normalized confusion matrices of RepSViT. Each column of the confusion matrix represents the predicted class, each row represents the true class, and the value in the grid represents the number of predicted categories of the corresponding column for the real category of the corresponding row.

Figs. 16 and 17 depict the confusion matrices of the different methods, allowing for a clear observation of the recognition performance for each ship class. The experimental results, as shown in Table III, indicate that RepSViT achieved the highest recognition accuracy of 95%. This demonstrates the outstanding object recognition capability of RepSViT. Additionally, RepSViT exhibits advantages in terms of parameters. With 3.77 M parameters, RepSViT has slightly more parameters than FastViT by 0.15 M and SOCNet, but still fewer parameters compared with other methods. Moreover, RepSViT demonstrates low computational costs, with only 0.6 GFLPs, which is slightly higher than EdgeViT [20] and SOCNet, but lower than the computational costs of other compared methods. Regarding inference speed, RepSViT achieves an inference latency of only 8.33 ms, which is higher than Spikingformer [22] and SOCNet [3] but lower than other compared methods. Although SOCNet has an advantage over RepSViT in terms of inference speed, computational costs, and parameters, RepSViT achieves higher recognition accuracy, striking a balance between speed and accuracy, thereby possessing higher practical applicability. As shown in Fig. 18, Latency, FLOPs, and accuracy visualization clearly shows that RepSViT has the highest accuracy when both computational expense and latency meet on-orbit requirements.

*2) Validation on the RSIOR-9 Dataset:* Table IV presents the experimental results of state-of-the-art

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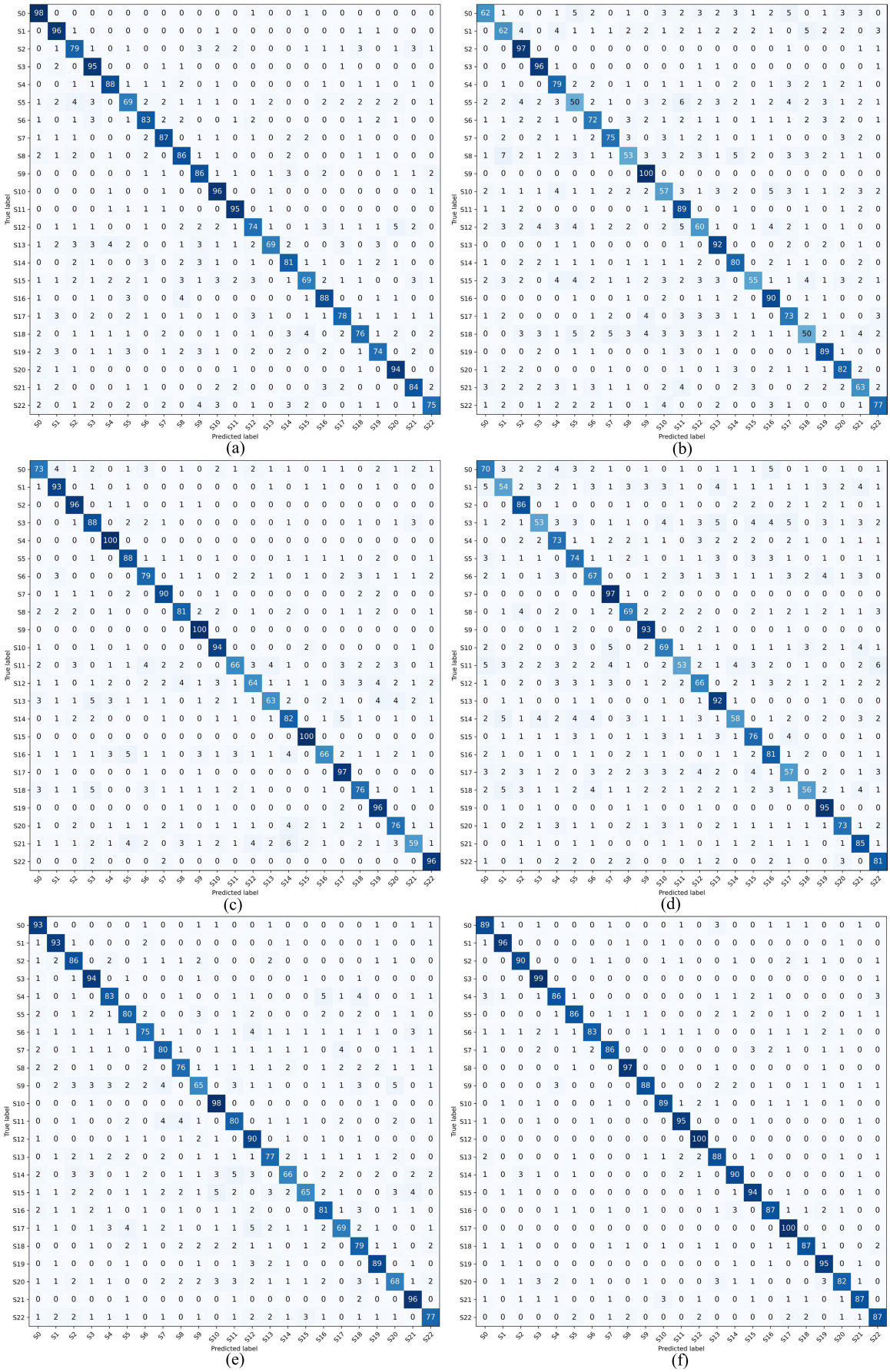



Fig. 17. Normalized confusion matrices of (a) FastViT, (b) LightViT, (c) EdgeViT, (d) Spikformer, (e) Spikingformer, and (f) SOCNet.

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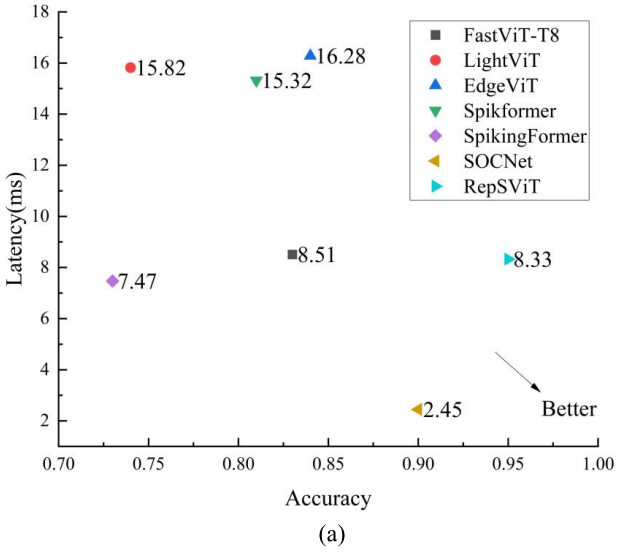



Fig. 18. Visual graphs for comparisons of the proposed methods on the FGSC-23 with other efficient ViTs and SOCNet. (a) Latency (ms)-accuracy scatter plot. (b) FLOPs (G)-accuracy scatter plot.

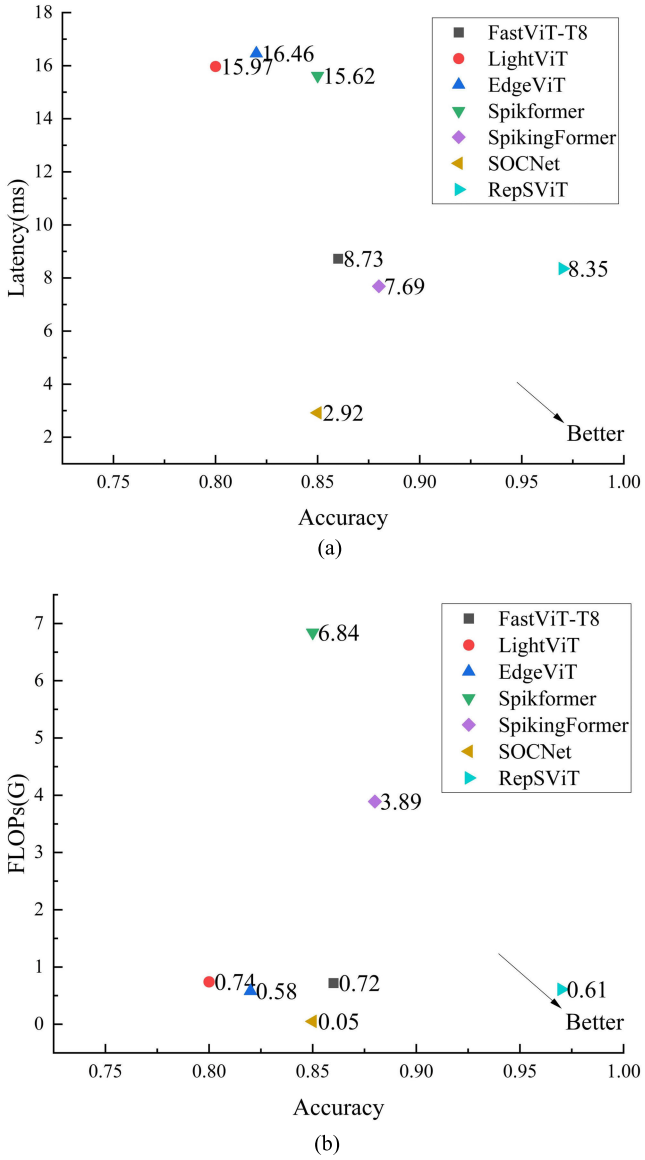



Fig. 19. Visual graphs for comparisons of the proposed methods on the RSIOR-9 with other efficient ViTs and SOCNet. (a) Latency (ms)-accuracy scatter plot. (b) FLOPs (G)-accuracy scatter plot.

TABLE IV

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE RSIOR-9

| Model | Latency(ms) | FLOPs(G) | Params(M) | Accuracy | Macro P | Macro R | Macro F1 |
|---|---|---|---|---|---|---|---|
| FastViT-T8 | 8.73 | 0.72 | 3.62 | 0.86 | 0.86 | 0.86 | 0.86 |
| LightViT | 15.97 | 0.74 | 9.40 | 0.80 | 0.79 | 0.80 | 0.79 |
| EdgeViT | 16.46 | 0.58 | 4.10 | 0.82 | 0.82 | 0.82 | 0.82 |
| Spikformer | 15.62 | 6.84 | 16.81 | 0.85 | 0.85 | 0.85 | 0.84 |
| SpikingFormer | 7.69 | 3.89 | 16.81 | 0.88 | 0.88 | 0.88 | 0.88 |
| SOCNet | **2.92** | **0.05** | **0.21** | 0.85 | 0.85 | 0.85 | 0.85 |
| RepSViT | 8.35 | 0.61 | 3.77 | **0.97** | **0.97** | **0.97** | **0.97** |

methods on RSIOR-9. It is evident from Table IV that RepSViT achieves the highest recognition accuracy of 0.97. Furthermore, RepSViT demonstrates an inference latency of 8.35 ms, ranking just below SOCNet and SpikingFormer. In terms of computational cost, RepSViT incurs 3.77M FLOPs, ranking second only to SOCNet, and FastViT-T8. Therefore, considering the processing environment in satellite on-orbit scenarios, RepSViT attains the highest recognition accuracy while maintaining competitive inference speed and computational cost, meeting the on-orbit requirements effectively. The visual representations in Fig. 19, illustrating comparisons between the proposed methods and other efficient ViTs and SOCNet on the RSIOR-9 dataset, further substantiate the aforementioned viewpoints. Table V provides a detailed account of precision, recall, and F1-score for each method across all categories. Superior performance metrics are highlighted in bold. It is evident from Table V that RepSViT exhibits the highest frequency of bolded entries across the three metrics, indicating its excellent recognition performance across various categories. Figs. 20 and 21 show the confusion matrix of the state-of-the-art methods and the confusion matrix of RepSViT, respectively, from which the specific recognition of each method for each category of objects can be seen, which further illustrates the excellent recognition performance of RepSViT across various categories.

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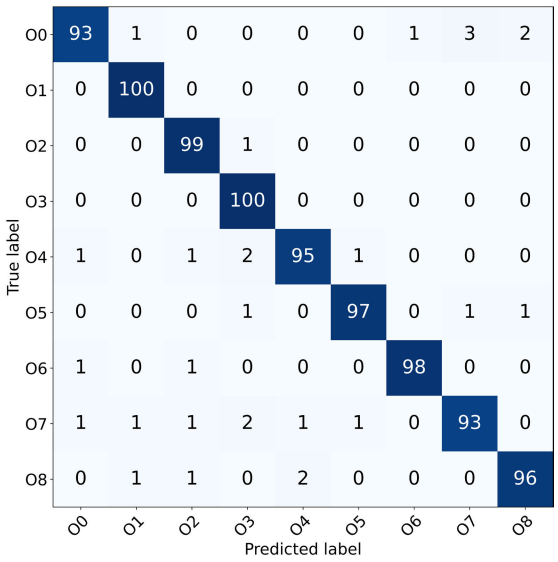



Fig. 20. Normalized confusion matrices of RepSViT.

TABLE V
DETAILED EXPERIMENTAL RESULTS OF ALL MODELS ON THE RSIOR-9

| | Category | O0 | O1 | O2 | O3 | O4 | O5 | O6 | O7 | O8 |
|---|---|---|---|---|---|---|---|---|---|---|
| FastViT | Precision | 0.86 | 0.86 | 0.87 | 0.82 | 0.88 | 0.89 | 0.87 | 0.88 | 0.83 |
| | Recall | 0.76 | 0.93 | 0.94 | 0.97 | 0.69 | 0.84 | 0.80 | 0.87 | 0.95 |
| | F1-Score | 0.81 | 0.89 | 0.90 | 0.89 | 0.78 | 0.87 | 0.83 | 0.87 | 0.88 |
| lightViT | Precision | 0.77 | 0.81 | 0.82 | 0.78 | 0.84 | 0.79 | 0.78 | 0.78 | 0.78 |
| | Recall | 0.81 | 0.85 | 0.97 | 0.75 | 0.93 | 0.63 | 0.93 | 0.60 | 0.69 |
| | F1-Score | 0.79 | 0.83 | 0.89 | 0.77 | 0.88 | 0.70 | 0.85 | 0.68 | 0.73 |
| EdgeViT | Precision | 0.84 | 0.80 | 0.85 | 0.86 | 0.79 | 0.81 | 0.84 | 0.76 | 0.84 |
| | Recall | 0.72 | 0.97 | 0.66 | 0.77 | 0.92 | 0.67 | 0.97 | 0.81 | 0.88 |
| | F1-Score | 0.77 | 0.88 | 0.74 | 0.81 | 0.85 | 0.73 | 0.90 | 0.79 | 0.86 |
| Spikformer | Precision | 0.85 | 0.84 | 0.87 | 0.78 | 0.92 | 0.84 | 0.84 | 0.84 | 0.87 |
| | Recall | 0.79 | 0.98 | 0.75 | 0.99 | 0.82 | 0.95 | 0.68 | 0.71 | 0.95 |
| | F1-Score | 0.82 | 0.90 | 0.81 | 0.87 | 0.87 | 0.89 | 0.75 | 0.77 | 0.91 |
| Spikingformer | Precision | 0.87 | 0.90 | 0.88 | 0.90 | 0.90 | 0.87 | 0.82 | 0.95 | 0.86 |
| | Recall | 1.00 | **1.00** | 0.94 | 0.95 | 0.95 | 0.72 | 0.77 | 0.70 | 0.91 |
| | F1-Score | 0.93 | 0.95 | 0.91 | 0.93 | 0.93 | 0.79 | 0.79 | 0.80 | 0.88 |
| SOCNet | Precision | 0.83 | 0.88 | 0.83 | 0.84 | 0.85 | 0.89 | 0.83 | 0.86 | 0.87 |
| | Recall | **0.96** | 0.76 | 0.96 | 0.83 | 0.88 | 0.78 | 0.85 | 0.78 | 0.84 |
| | F1-Score | 0.89 | 0.82 | 0.89 | 0.83 | 0.86 | 0.83 | 0.84 | 0.82 | 0.85 |
| RepSViT | Precision | **0.97** | **0.97** | **0.96** | **0.94** | **0.97** | **0.98** | **0.99** | **0.96** | **0.97** |
| | Recall | 0.93 | **1.00** | **0.99** | **1.00** | **0.95** | **0.97** | **0.98** | **0.93** | **0.96** |
| | F1-Score | **0.95** | **0.99** | **0.98** | **0.97** | **0.96** | **0.97** | **0.98** | **0.94** | **0.96** |

## V. CONCLUSION

This article presents an object recognition model, RepSViT, for satellite on-orbit computing. The model introduces SNNs into the field of satellite on-orbit computing for the first time while utilizing structural reparameterization techniques. Specifically, we design a $D^2SC$ based on SNNs to improve the feature extraction capability and efficiency of RepSViT. We also develop an SGAM to make RepSViT pay more attention to object-related features with lower computational costs. Moreover, ECFC is introduced to enhance the model's coarse-fine-grained feature extraction capability. To ensure feature extraction ability, inference speed, and efficiency, we propose RepFFN. The proposed model undergoes ablation experiments and comparison experiments with state-of-the-art methods on the publicly available FGSC-23 and RSIOR-9. RepSViT achieves an inference latency of 8.33 ms and a recognition accuracy of 95% on an embedded GPU with FGSC-23 dataset, utilizing 3.77 million parameters and consuming 0.6 GFLOPs computational costs. RepSViT achieves an inference latency of 8.35 ms and a recognition accuracy of 97% on an embedded GPU with RSIOR-9 dataset, utilizing 3.77 million parameters and consuming 0.61 GFLOPs computational costs.

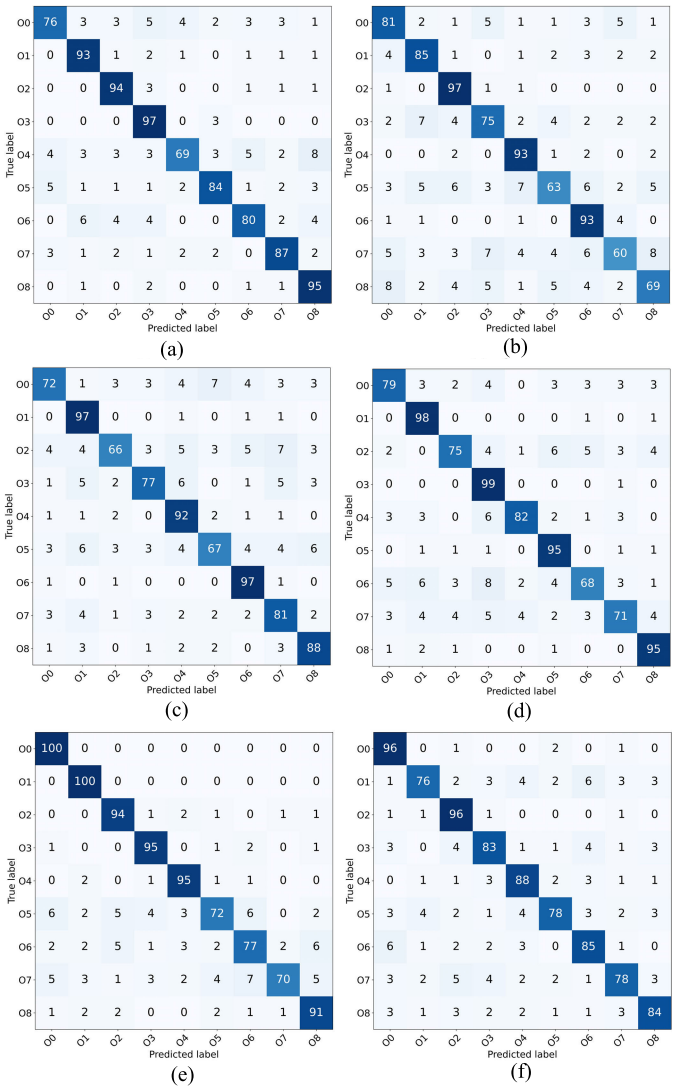



Fig. 21. Normalized confusion matrices of (a) FastViT, (b) LightViT, (c) EdgeViT, (d) Spikformer, (e) Spikingformer, and (f) SOCNet.

The experimental results demonstrate the effectiveness and advancement of the proposed model. RepSViT is capable of adapting to extremely limited computing resources on satellites while striking a tradeoff between computational costs, inference speed, and inference accuracy, providing a new solution for on-orbit object recognition tasks in satellites.

Due to the unavailability of neuromorphic hardware at present, this article presents only the experimental results of RepSViT on GPU. In the future, we will explore its performance on neuromorphic hardware, and we believe neuromorphic hardware will further improve results by reducing the number of operations through model sparsity, simultaneously reducing the computational costs of GFLOPs.

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